

CODE A DANCE PARTY!

Choreographing with Scratch Jr

Hey Magical Code Dancers! 🌟

Have you ever danced at a school party, a festive wedding, or celebrated a fun festival like Diwali or Navratri? The amazing rhythm, your stepping, and your spins are just like writing code for a computer! Today, you are the master dance choreographer! We will use simple coding blocks to create an epic, animated dance party! 🎉

What is a Choreographer?

Before you perform on a big stage, a dance teacher shows you exactly when to step left, when to clap your hands, and when to spin around. That teacher is a **Choreographer**! Today, YOU are the choreographer, but instead of training humans, you will code cute digital characters to dance perfectly to your beats! 🤖💃

Our Secret Coding Translation Card

Programmers sometimes use giant words, but you already know what they mean from the playground! Let's translate them together:

Big Word	What it Actually Means	Fun Dance Example
Linear Script	Doing steps in a straight line, one after another.	First you bow, then you jump, then you smile! 😊
Sequence	The perfect order of your steps. Don't mix them up!	Putting your shoes on <i>before</i> you tie the laces! 👟
Parameter		

A number input that changes how big or fast an action is.

Taking **5** small hops instead of just **1** giant leap! 🐾

Loop

A magic box that tells a step to repeat over and over again.

Clapping your hands **4** times to match a drum beat! 🥁

Warm-Up Game: The Human Code Machine!

Let's stand up and turn our classroom into a live, physical computer network before we touch our screens!

How to Play:

- **Form Code Duos:** Team up with a partner! One is the *Software Programmer* and the other is the *Mechanical Robot Sprite*.
- **Hold Up Flashcards:** The Programmer holds up colorful action cards like **MOVE FORWARD 2 STEPS** or **REPEAT 3 TIMES**.
- **Strict Robot Rules:** The Robot Sprite must do these exact actions from start to finish without making any changes!

Dance Style 1: The High-Energy Bhangra!

Our first dance comes from the sunny green fields of Punjab. It is filled with giant smiles, huge hops, and arms pointing high up to the sky! ✨



Physical Step Guide:

Stand up straight. Raise both hands up in the air above your shoulders. Now, bounce your right knee up high while pulsing your hands up to the sky, then switch to your left knee! It feels like you are bouncing on a fun trampoline!

How We Build Bhangra with Scratch Jr Blocks:

Scratch Jr Block	Block Group	What It Tells the Actor to Do
Green Flag	Yellow Triggers	Signals the dance to start immediately when you click Go!

Hop (Up & Down)	Blue Motions	Commands our character to leap high into the air and land back down.
Grow / Shrink	Purple Looks	Changes the character's size parameters, making them bounce rhythmically!

Dance Style 2: The Swirling Garba Spin!

Our next dance travels to the festive streets of Gujarat. People gather in giant, beautiful circles to dance together under the stars during Navratri! 🌟

Physical Step Guide:

Imagine walking along a giant circular track. Take three graceful steps forward, clap your hands together tightly to make a loud sound, and then do a full, swirling circle spin right on your tiptoes!

The Garba Logic Block Formula:

- **The Walk Input:** Use the **Move Right** block and set the number parameter to **4**.
- **The Festive Sound:** Attach the purple **Play Pop Sound** block to mimic the hand clap rhythm!
- **The Big Twirl:** Snap in the circular **Turn Clockwise** arrow and type in the number **12** to make a full, beautiful spin!

Setting Up Your Digital Dance Stage

Now open your tablet, launch the **Scratch Jr** app, and set up your stage!

STEP 1: BUILDING THE THEATER BACKDROP

Click on the landscape picture icon at the top header bar. Look for the **Stage** backdrop (with the gorgeous red velvet curtains) or the festive **City Park at Night** setting!

STEP 2: CASTING YOUR DANCE TROUPE ACTORS

Go to your character library. Delete the default cat by holding your finger down on it until it wiggles and shows an 'X'. Tap the big plus (+) button twice to add two custom performers:

- **Performer 1 (Bhangra):** Drag them to the left side of the stage.
- **Performer 2 (Garba):** Drag them to the right side of the stage.

 *Creative Idea: If you want, you can code a happy **Elephant** to do the Bhangra or a cute **Frog** to spin the Garba! Let your imagination run wild!*

Assembling the Code Trains

Click on your performers and snap your puzzle blocks together tightly from left to right in the lower canvas workspace area:

The Bhangra Code Recipe Guide

1. **Green Flag Block:** Wakes up our dancer the millisecond the main green flag is touched!
2. **Hop Block (Set to 2):** Tells our dancer to bounce up into the air!
3. **Grow Block (Set to 2):** Makes him grow bigger to look like he jumped forward!
4. **Shrink Block (Set to 2):** Resets his size back to normal.
5. **Forever Loop Arrow:** Wrap this bright orange arrow all the way around the hop and size blocks so he keeps grooving endlessly!

Coding Tip: Make sure Grow and Shrink use the exact same number. If they don't, your dancer will keep growing until he fills the whole screen like a giant monster! 🤖

The Garba Code Recipe Guide

1. **Green Flag Trigger:** Starts her performance in perfect timing with her partner.
2. **Move Right (Set to 3):** Teaches her to slide gracefully across the stage.
3. **Play Pop Sound:** Triggers a sharp audio snap that mimics her clapping!
4. **Turn Clockwise (Set to 12):** Commands her to spin all the way around on her tiptoes.
5. **Forever Loop:** Locks the entire line in a continuous loop so she never stops her graceful dance.

Becoming a Professional Dance Doctor! (Debugging)

In computer science, a code mistake is called a **Bug** , and finding it and fixing it is called **Debugging**! Let's learn how to squash two common dance bugs:

BUG CASE #1: THE GIANT GROWING PERFORMER

The Sad Sight: Your dancer starts hopping, but every time he lands, he grows bigger and bigger until his giant head blocks out the scenery!

The Easy Fix: Check your parameters! This happens when you add a purple Grow block but forget to add the matching Shrink block right after it. Remember, what grows big must shrink back to normal size!

BUG CASE #2: THE DIZZY LEANING PERFORMER

The Sad Sight: Your dancer starts spinning, but instead of finishing upright, she stops halfway through and stays tilted sideways like she is falling over!

The Easy Fix: In Scratch Jr, it takes a numerical parameter value of **12** to make a perfect, complete circular turn. If you typed in a small number like 3 or 6, she gets stuck. Change that number to **12**!

The Dance Showcase Trivia Game!

QUESTION 1: THE FAST TEMPO MIX-UP

Your dance teacher tells you the music is speeding up fast! You need your character to hop up and down much quicker. What should you adjust?

☐ Option A: Swap the backdrop picture.

☒ **Option B: Change the number parameter value on the blue Hop block. (Correct!)**

☐ Option C: Change the color of the character's dress.

QUESTION 2: THE ONE-TIME DANCER CHALLENGE

You want your dancer to perform her routine just **one single time** and then stop and bow, instead of dancing forever. What block should you remove?

☒ **Option A: Remove the orange Forever Loop block. (Correct!)**

☐ Option B: Remove the yellow Green Flag block.

Your Take-Home Code Mission: Family Grooves!

Congratulations, Master Digital Choreographers! 🎉 You have officially graduated from our Dance Party Lab.



My Personal Dance Coding Log Sheet

1. Name of My Family Heritage Dance Style:

2. What blocks did I use for this new dance style? (Check the boxes below):

☐ Green Flag ☐ Move Left/Right ☐ Turn Spin ☐ Forever Loop

3. Did my family test and play my dance game?

☐ YES! ☐ NOT YET!